2026-09-11

The Blind Spot Paradox

The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

Alexandre Boyer Mélaine Gouillou Salomé Fonvielle Ulysse Petit-Tichanné

Research Track, Topic 39, CentraleSupélec. Supervisor: Raphaël Minato

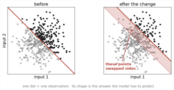
The Blind Spot Paradox: when a self-repairing model destroys the evidence its own monitor needs.

2026-09-11

The Blind Spot Paradox

└─Concept drift: the rule changes, the data does not

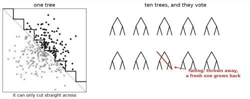
Concept drift: the rule changes, the data does not



A model trained on the left keeps predicting, confidently, and is now wrong on the shaded band.

First, the phenomenon. A model learns a rule: this input goes with that answer. Then the world changes and the rule no longer holds. That is concept drift, and here is the cleanest version of it, the one we simulate. Two inputs, a straight boundary, a label on each side. The change moves the boundary, and nothing else. Only the shaded band has swapped sides, and a model trained before is now wrong on it.

Defence one: a model that repairs itself

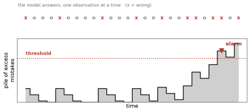


This is the Adaptive Random Forest. Ten trees, replaced one at a time.

The first defence repairs itself. Its building block is a decision tree: it cuts the picture straight across, so a slanted boundary can only be approximated by a staircase. One tree is fragile, so ten of them vote. That is the Adaptive Random Forest, the model we study. A tree whose answers go wrong is thrown away, and a fresh one grows in its place. Ten trees, replaced one at a time.

Defence two: a watchdog that raises the alarm

Defence two: a watchdog that raises the alarm



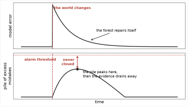
Two communities, two tools, never measured against each other.

The second defence comes from a different community. A watchdog keeps a pile of the model's excess mistakes. While the model does well the pile falls back to zero; let the mistakes come and it grows, until it crosses a threshold and calls a human. That alarm is the only thing a human ever sees. Two communities, two tools, never measured against each other.

The Blind Spot Paradox

└─Put them together, and they cancel out

Put them together, and they cancel out



schematic

Put the two together, and they cancel out. The boundary moves, the mistakes begin, and the pile starts climbing. But the forest repairs itself, the mistakes stop, and the pile drains back to zero without ever reaching the threshold. The repair erased the evidence. The better the model fixes itself, the more surely the alarm stays silent.

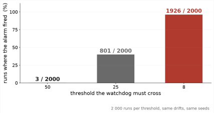
2026-09-11

The Blind Spot Paradox

└─Why not just lower the threshold?

Does the alarm ever fire? That depends on the threshold. We re-ran the benchmark at three settings, two thousand runs each. At the highest it fired three times out of two thousand. Lower it, and it fires almost every time. So why not lower it? Because a low threshold also fires when nothing happened, and an alarm that cries wolf gets switched off. The setting quiet enough to be trusted is the one that never sees the drift.

Why not just lower the threshold?



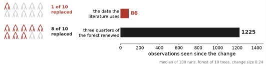
A low threshold also fires when nothing has happened, and an alarm that cries wolf gets switched off.

2026-09-11

The Blind Spot Paradox

└─Our contribution: when is the forest actually repaired?

Our contribution: when is the forest actually repaired?



The accepted date fires at the *first* tree. It times the start of a repair, not its end.

It is a race, so we have to time both runners. Timing the alarm is easy. Timing the repair is where we found something. The literature has a standard date for it, and that date turns out to be the moment the very first tree out of ten is replaced. Wait until three quarters of the forest has actually been renewed, and you wait roughly ten times longer. That stopwatch times the start of a repair, not its end, and it is wrong always in the same direction.

Where we stand

- The paradox is real and reproducible, and we can say when the evidence falls short of what the alarm needs.
- The clock the field relies on is biased, always the same way. Every conclusion drawn from it inherits that bias.
- **Next**: watch the model and the watchdog on one common clock, and turn this into advice a practitioner can use.

Thank you.

Where we stand. The paradox is real and reproducible. The clock the field relies on is biased, always the same way, so every conclusion drawn from it inherits that bias. What remains is to turn this into advice a practitioner can use. Thank you.